

Introduction

Baroid Core Homework 7

Assessment Feedback

Congratulations, you passed.

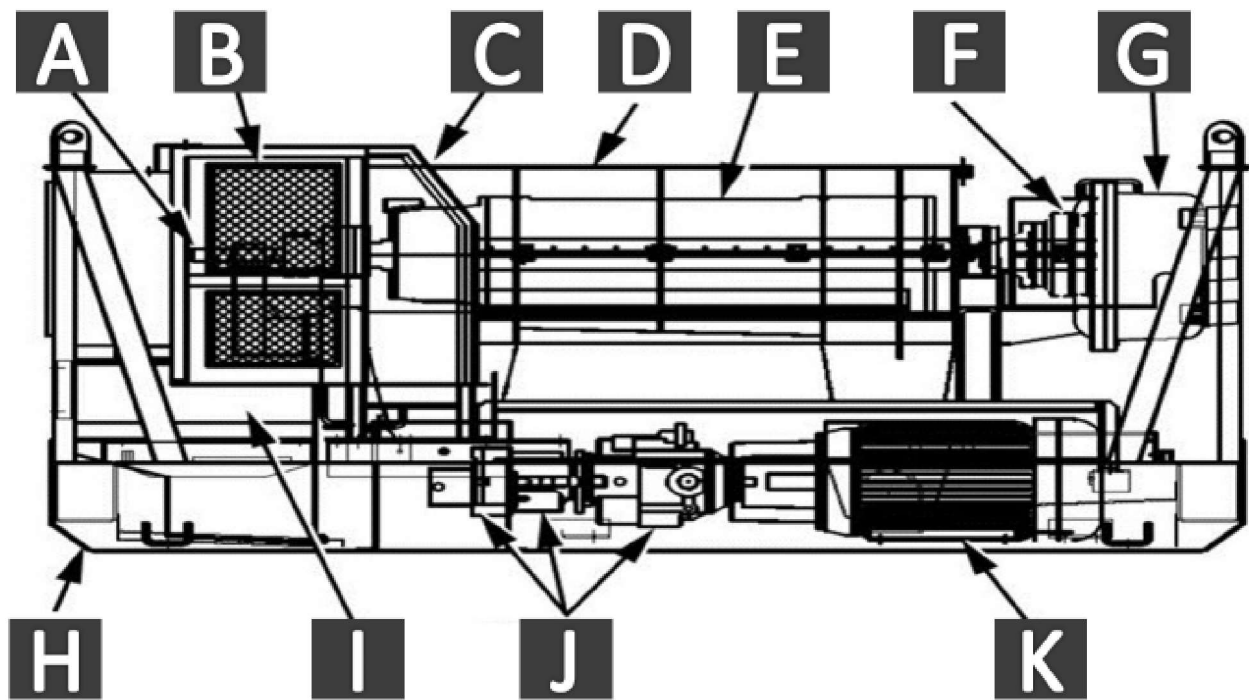
Total score: 138 out of 139, 99%

Question Feedback

Question

1 of 48

Label the major components of a Dual Voltage DE-1000 FHD using the diagram below:



- | | |
|--------------------------|--------------------------------|
| Electrical control panel | <input type="text" value="G"/> |
| Skid | <input type="text" value="H"/> |
| Case | <input type="text" value="D"/> |
| Heat Exchanger | <input type="text" value="B"/> |

Hydraulic Pumps	J ▾
Feed Tube	A ▾
Rotating Assembly	E ▾
Hydraulic Control Panel	C ▾
Centrifugue Motor	K ▾
Reservoir	I ▾
Rotodiff	F ▾

11 out of 11

Question

2 of 48

The replaceable wear parts of a Dual Voltage DE-1000 FHD are (select all that apply):

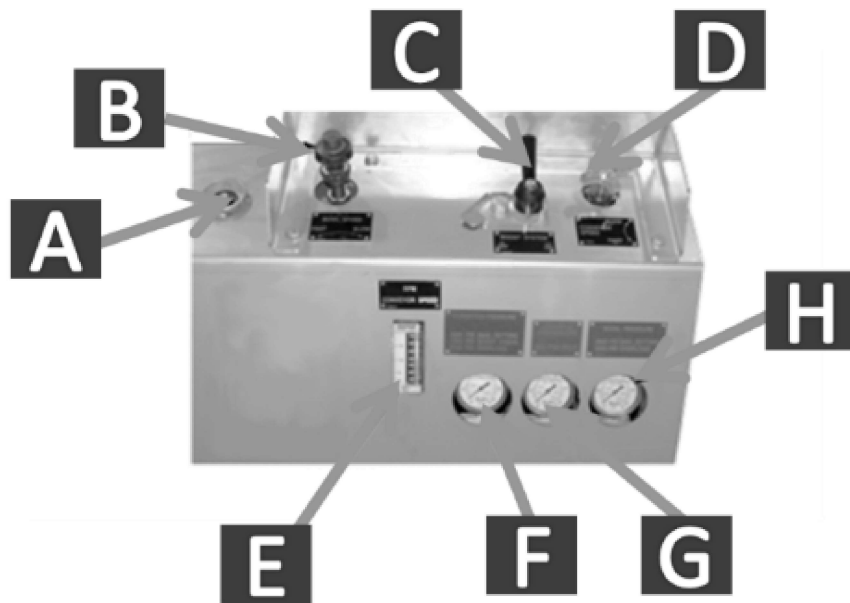
- ☒ Feed accelerator
- ☐ Pillow blocks
- ☒ Feed nozzle liners
- ☒ Solids discharge wear inserts
- ☒ Feed tube

4 out of 4

Question

3 of 48

Using the diagram below, identify the controls, gauges, and indicators on the Dual Voltage DE-1000



- | | |
|--------------------------|-----|
| Conveyor speed indicator | E ▾ |
| Boost system switch | C ▾ |
| Charge pressure gauge | G ▾ |
| Conveyor speed Control | D ▾ |
| Clogged filter indicator | A ▾ |
| Bowl speed control | B ▾ |
| Conveyor pressure gauge | F ▾ |
| Bowl pressure gauge | H ▾ |

8 out of 8

Question

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The feed pump on a Dual Voltage DE-1000 FHD must be connected to the Electrical control panel ▾ for feed pump cycling to function automatically.

1 out of 1

Question

5 of 48

During feed pump cycling on a Dual Voltage DE-1000 FHD, the conveyor pressure stops the feed pump at 1500 psi ▾ and restarts the feed pump at 700 psi ▾

2 out of 2

Question

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The boost system on a Dual Voltage DE-1000 FHD increases conveyor speed when conveyor pressure reaches 1600 psi ▼ and the centrifuge shuts down when conveyor pressure reaches 2100 psi ▼

2 out of 2

Question

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All the effluent ports on a centrifuge set must be set in the same position

- ☒ True
- ☐ False

1 out of 1

Question

8 of 48

What covers and guards have proximity sensors on a Dual Voltage DE-1000 FHD (select all that apply)?

- ☒ Case cover
- ☐ Feed pump starter box case
- ☒ Rotodiff cover
- ☒ Belt guard
- ☐ Electrical control panel case

3 out of 3

Question

9 of 48

What is the bowl diameter of a DE-7200?

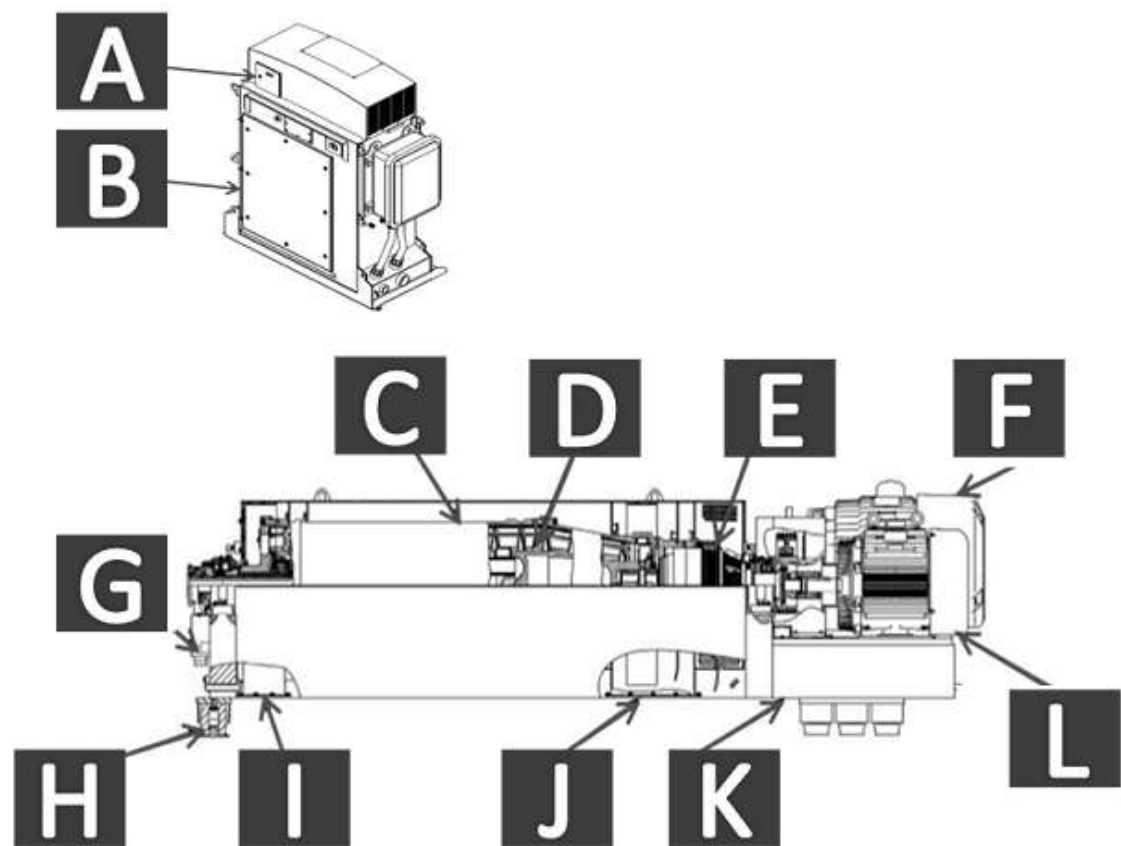
- ☐ 17 inch
- ☐ 18.5 inch
- ☐ 20 inch
- ☒ 21.5 inch
- ☐ 23 inch

1 out of 1

Question

10 of 48

Label the major components of a DE-7200 on the diagram below



- | | |
|--------------------------|-----|
| Control cabinet | B ▾ |
| Vibration isolator mount | H ▾ |
| Conveyor drive motor | L ▾ |
| Case and base | K ▾ |
| Bowl drive motor | F ▾ |
| Purge Unit | A ▾ |
| Vibration switch | G ▾ |
| Conveyor drive gearbox | E ▾ |
| Liquid discharge chute | I ▾ |
| Solid discharge chute | J ▾ |
| Conveyor | D ▾ |
| Bowl | C ▾ |

12 out of 12

Question

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What are the causes of automatic shutdown on the DE-7200?

- ☐ Excessive Vibration
- ☐ Main Bearing Temperatures
- ☐ Excessive Bowl Speed

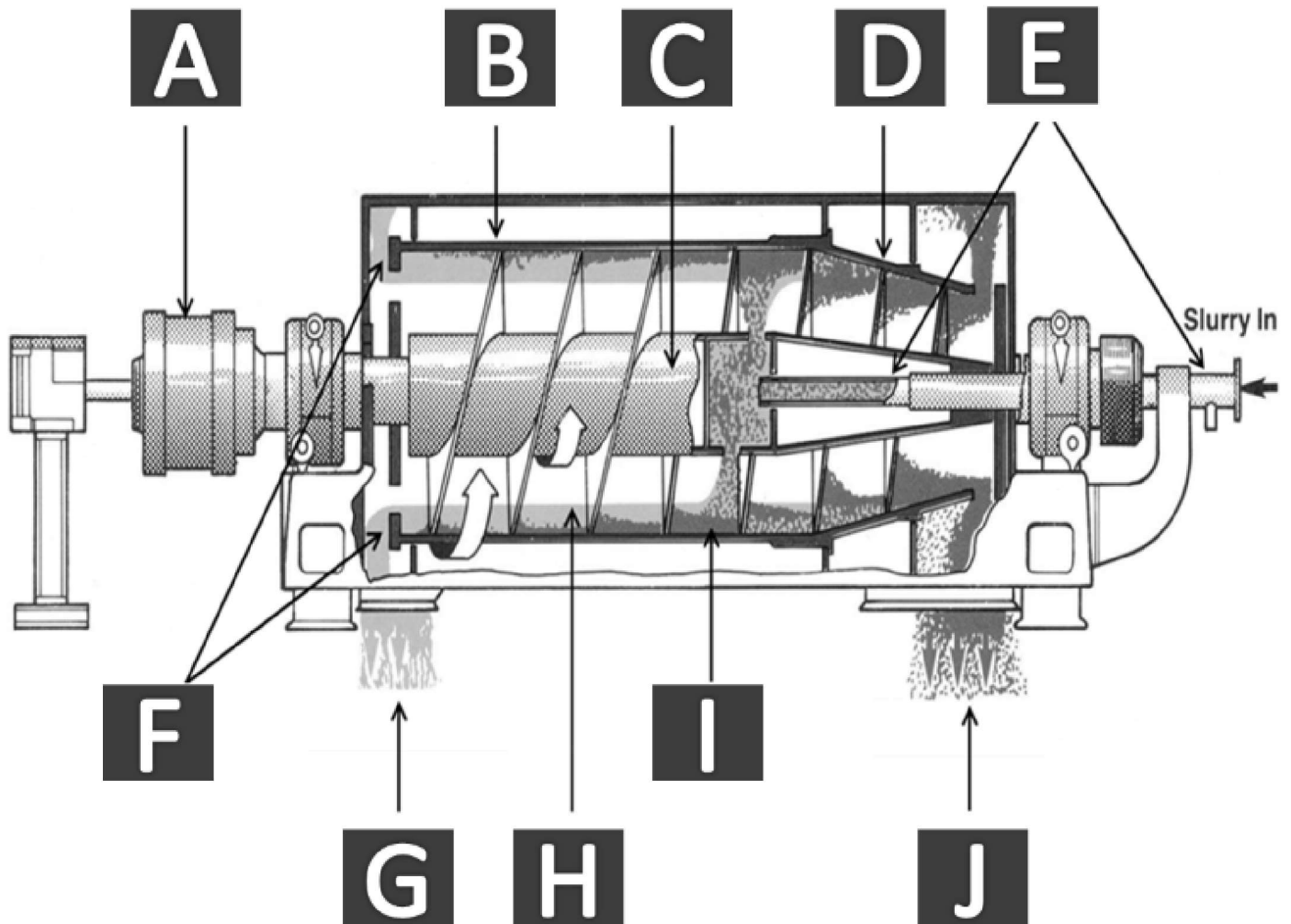
- ☐ Conveyor Clutch Trip
- ☒ All of the above

1 out of 1

Question

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Label the major parts of a centrifuge, using the diagram below:



Pond
Bowl
Conveyor
Overflow (effluent)
Wall cake
Feed
Gear Box
Weirs

H ▾
B ▾
C ▾
G ▾
I ▾
E ▾
A ▾
F ▾

Underflow (solids discharge) J ▾

Beach D ▾

10 out of 10

Question

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Centrifuge performance is defined by: _____

- ☐ Processing capacity
- ☐ Dryness of solids discarded
- ☐ Density reduction
- ☐ Quantity of solids removed
- ☐ Size of solids removed
- ☒ All of the above

1 out of 1

Question

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The performance factors of a centrifuge with regards to the feed are: _____

- ☐ Quantity and size distribution of total solids
- ☐ Properties of individual solids
- ☐ Feed rate
- ☐ Drilling fluid properties
- ☒ All of the above

1 out of 1

Question

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The performance factors of a centrifuge with regards to the centrifuge itself are: _____

- ☒ Bowl geometry, centrifugal forces, retention time in the pond, dewatering time on the beach, and torque
- ☐ Bowl motor size, conveyor motor size, feed tube length, and type of feed pump
- ☐ Feet of head from pump, pressure at feed tube, type of conveyor
- ☐ Height of the centrifuge stand and length of hose from the feed pump

1 out of 1

Question

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From the following operational variables of a centrifuge that can be adjusted, indicate what operational parameter will be altered by the given adjustment

Feed rate	Retention time and active processing % ▼
Pond depth	Retention time and dewatering time ▼
Feed dilution	Particle settling velocity ▼
Bowl speed	G-force ▼
Conveyor speed	Dewatering time and torque ▼

5 out of 5

Question

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When working with a centrifuge, if the fluid density increases, the settling velocity increases and the cut point is coarser

- ☐ True
☒ False

1 out of 1

Question

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What are the G's of a centrifuge with a 14 inch bowl operating at 2000 RPM?

Use the formula below to calculate

$$Gforce = \frac{Bowl\ diameter\ (inches) \times (RPM)^2}{70414}$$

G force (2 decimals rounding) 795.3

3 out of 3

Question

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Using the results from the previous question, calculate at what speed (RPM) must a 21.5 inch bowl centrifuge operate to achieve the same G force?

RPM (2 decimal rounding) = 1613.9

3 out of 3

Question

20 of 48

The three primary applications for centrifuges utilized by BSS during drilling operations are: _____

- ☐ Weight up the active drilling fluid
- ☒ Max solids removal from un-weighted mud
- ☒ Control rheology of weighted mud
- ☒ Chemically enhanced centrifugation
- ☐ Secondary drying
- ☐ Separate barite from drilled solids

3 out of 3

Question

21 of 48

Complete the analysis below, identifying changes to centrifuge parameters required to achieve finer cut point and dryer discharge, when using a centrifuge on un-weighted mud for solids removal.

- | | |
|---|---|
| If the Bowl speed initial setting is 1500+ G's. | Decrease to reduce torque, Increase finer cut point, Increase dryer discharge ▼ |
| If Conveyor speed initial setting is moderate RPM | Increase as needed to reduce torque, Decrease finer cut point, Decrease dryer discharge ▼ |
| If Pond depth initial setting is medium height | Decrease, Increase finer cut point, Decrease dryer discharge ▼ |
| If Feed rate initial setting is moderate GPM | Increase, Decrease finer cut point ▼ |

4 out of 4

Question

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When using a centrifuge, if the desired effect is to remove finer solids, the bowl RPM should be increased and the pond depth should increase as well

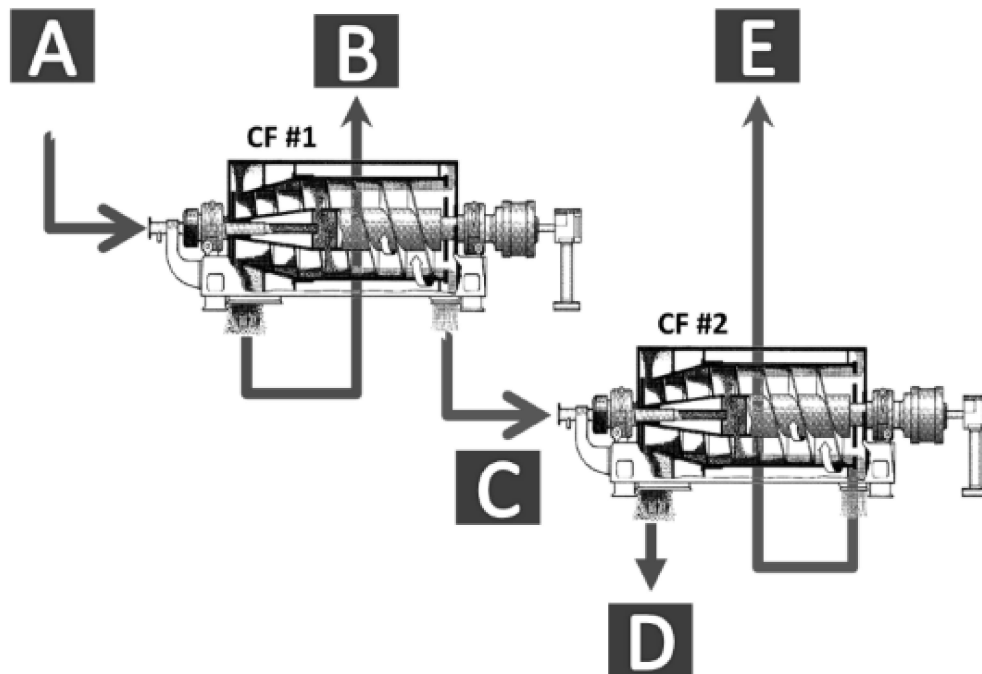
- ☒ True
- ☐ False

1 out of 1

Question

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On the diagram below, identify the material flows for two-stage centrifuging on weighted mud to control rheology



HGS to active system

B ▼

Liquid Phase to the active system

E ▼

Feed from active system

A ▼

LGS to discard

D ▼

LGS + Liquid phase move to second centrifuge

C ▼

5 out of 5

Question

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In regards to the previous diagram, **select all statements** below that best describe the reasons for the specified bowl and conveyor settings on the first centrifuge (CF #1)

- ☒ High conveyor speed is utilized to ensure discharge to active system remains wet
- ☐ Low Conveyor speed is utilized to produce dry solids returned to the active system
- ☒ Low bowl speed is utilized to minimize settling / removal of LGS discharged to the active
- ☐ High bowl speed is utilized to remove all solids to be returned to the active system

2 out of 2

Question

25 of 48

Explain why a LOW or HIGH pond depth would **NOT** be acceptable pond depth setting for the first centrifuge (CF#1).a) Too HIGH pond depth affects negatively the Increased Retention Time ▼ and the result would be: LGS settling ▼ and returned to active system

Too LOW pond depth affects what parameter negatively: Increased Dewatering Time ▼ and the result would be: Consistency of returned HGS is too dry & doesn't mix ▼

3 out of 4

Question

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Provide two reasons why a “big bowl” centrifuge would be the most appropriate choice if needing to improve the removal of fine LGS

- ☒ Increased retention time
- ☐ Lower volume of fines
- ☐ Increased dewatering time
- ☒ Increased G-force
- ☐ All of the above
- ☐ None of the above

2 out of 2

Question

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List the three different types of centrifuges (In regards to drive type):

- ☒ Full Hydraulic Drive
- ☒ Gear Box Drive
- ☐ Fully automated Drive
- ☒ Variable Frequency Drive
- ☐ Hybrid box drive

3 out of 3

Question

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The Dewatering process achieved with BaraH₂O is often referred to as:

- ☒ Chemically enhanced Waste Reducer
- ☐ Dissolved Air flotation
- ☐ Liquid clarification
- ☒ Chemically enhanced solid removal

Question

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When dewatering, **Coagulants** ▼ are used to neutralizes the negative charge of the fine suspended particles allowing the solid particles to form mini aggregates

1 out of 1

Question

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When dewatering, **Flocculants** ▼ are used to envelope the aggregated solids to form larger solid clusters called flocs.

1 out of 1

Question

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Once the mud has been chemically treated in the Enviro-Floc unit it is pumped to a **Centrifuge** ▼ for the flocculated solid particles to be mechanically removed

1 out of 1

Question

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Select the two types of cutting drying technologies most commonly utilized by BSS:

- ☒ Vertical Cuttings Dryer
- ☒ Secondary Drying Shaker
- ☐ Horizontal Cuttings Dryer
- ☐ Rotary Vacuum Dryer
- ☐ Thermal Phase Separator
- ☐ High Velocity Oil Retainer

2 out of 2

Question

33 of 48

Choose the correct method(s) of solid / fluid separation utilized by a Vertical Cuttings Dryer:

- ☐ Evaporation
- ☒ Centrifugal Force
- ☐ Vacuum reduction

- ☒ Filtration
- ☐ Gravitational settling

2 out of 2

Question

34 of 48

What model of Vertical cutting Dryers does BSS offer?

- ☒ V-71
- ☐ V-77
- ☒ V-133
- ☐ V-311

2 out of 2

Question

35 of 48

What is the range G-force exerted by a vertical dryer?

- ☒ 500-700 G's
- ☐ 300-500 G's
- ☐ 1200+ G's
- ☐ 200 G's

1 out of 1

Question

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Which pieces of equipment besides the vertical dryer and control panel, are needed to complete the vertical dryer operations? (Circle all that apply)

- ☐ Hydrocyclone
- ☐ Drying Shaker
- ☒ Recovery Tank
- ☒ Centrifuge
- ☒ Pump
- ☒ Discharge System
- ☐ Dewatering Unit
- ☒ Delivery System

5 out of 5

Question

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What are the two main components of the vertical dryers rotating assembly?

- ☐ Rotating slotted drum
- ☒ Rotating wedged wire basket
- ☒ Rotating scroll with vertical flights
- ☐ Rotating scroll with horizontal flights

2 out of 2

Question

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Is the vertical dryer considered to be a “stand alone” solid control device?

- ☐ Yes
- ☒ No

1 out of 1

Question

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What statements are **NOT** true about the vertical dryer operations?

- ☒ The vertical Dryer is considered a solids control device
- ☒ Fluid recovered in the collection tank require no further processing
- ☐ Reduces haul-off and disposal costs
- ☐ Involves dangerous elements of operation
- ☐ Meets EPA requirements for ROC discharge limits in certain areas
- ☒ Requires very little attention and maintenance

3 out of 3

Question

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What is the typical ROC % achieved at the discharge of the vertical cuttings dryer?

- ☐ <2%
- ☐ <3%
- ☐ <4%
- ☒ <5%
- ☐ None of the above

Question

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What is the optimum flight to screen clearance for the vertical dryer? _____

- ☐ 1"
☒ 0.020"
☐ 0.5"
☐ 0.05"

1 out of 1

Question

42 of 48

How much new volume (dilution) is required to maintain 6% drill solids for 3500' of 13 7/8" hole with 80% solids removal efficiency?

Use the formulas provided in class (see below):

SRE

1. $\frac{ID^2}{1029.4} \times \text{Feet drilled} = \text{Total Drilled Solids (TDS) generated, bbl}$
2. $\text{TDS} \times \text{Decimal fraction of drill solids remaining in the system} = \text{Drilled Solids Remaining (DSR) in system, bbl}$
3. $\frac{(\text{DSR}) \text{ Drilled Solids Remaining}}{\text{Decimal \% of Solids to maintain}} = \text{Total Solids \& Liquid (TSL) in system to maintain \% solids, bbl}$
4. $(\text{TSL}) \text{ total solids \& liquid} - (\text{DSR}) = \text{Dilution Required}$

Note:

- When we use % in an equation we change % into a decimal fraction
- We assume we drill a gauge hole.

Needed dilution volume in bbl = 2050.92

3 out of 3

Question

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How much new volume (dilution) is required to maintain 6% drill solids for 3500' of 13 7/8" hole with 92% solids removal?

Use the formulas provided in class (see question above)

Needed dilution volume in bbl = 820.31

3 out of 3

Question

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If each barrel that you saved cost \$150/barrel, what is the cost savings for Question 42 vs. Question 43?

Cost savings in US\$ = (type whole number without commas or decimals) 184592

3 out of 3

Question

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75% SRE with 8% drill solids for 1500' of 12 1/4" hole. What is additional dilution volume?

Needed dilution volume in bbl = 628.71

3 out of 3

Question

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90% Solids Removal Efficiency with 8% drill solids for 1500' of 12 1/4" hole. What is additional dilution volume?

Needed dilution volume = 251.51

3 out of 3

Question

47 of 48

If each barrel that you saved cost \$180/barrel, what is the cost savings for Question 45 vs. Question 46?

Cost savings US\$ = (Type whole number without commas or decimals) 67896

3 out of 3

Question

48 of 48

Explain why a LOW or HIGH pond depth would NOT be acceptable pond depth setting for the first centrifuge (CF#1) on the diagram above. (Select all applicable answers)

- ☒ Too LOW pond depth affects Increased Dewatering Time
- ☐ Too LOW pond depth affects Increased Retention Time
- ☒ Too HIGH pond depth affects negatively the Increased Retention Time
- ☐ Too HIGH pond depth affects negatively the Increased Dewatering Time

2 out of 2